

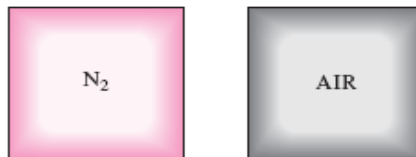
FUNDAMENTALS OF THERMODYNAMICS

Contents:

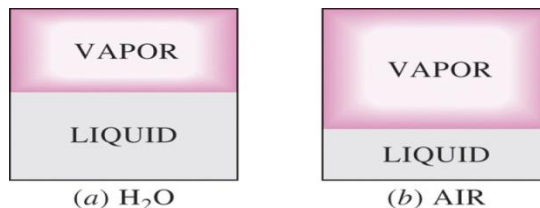
- Concepts of pure substance, types of systems , properties of systems- Extensive and Intensive properties, processes and cycles, Quasi-static process, flow and non flow process, Thermodynamic equilibrium, Point and path function. **04 Marks**
- Work, Heat Transfer and Energy. Thermodynamic definition of work & heat, Difference between heat and work. Energy –Potential Energy, Kinetic Energy, Internal Energy, Flow Work, concepts of enthalpy & entropy. **04 Marks**
- Laws of Thermodynamics- Zeroth Law, principle of law of conservation of energy First law of Thermodynamics, Second Law of Thermodynamics- Kelvin Planks, Clausius statements and their equivalence, Clausius inequality, Concept of perpetual motion machine of first and second kind. **06**
- Application of Laws of Thermodynamic:- Steady Flow Energy equation and its application to boilers, engine, nozzle, turbine, compressor & condenser. Application of Second law of Thermodynamics to Heat Engine, Heat Pump and Refrigerator. **06 Marks**

❖ Concepts of Pure Substance:

- A substance that has a fixed chemical composition throughout is called a pure substance.



- A pure substance does not have to be of a single chemical element or compound, however. A mixture of various chemical elements or compounds also qualifies as a pure substance as long as the mixture is homogeneous.
- A mixture of liquid and water vapor is a pure substance, but a mixture of liquid and gaseous air is not.



- Examples:
 1. Water (solid, liquid, and vapor phases)
 2. Mixture of liquid water and water vapor
 3. Carbon dioxide, CO₂
 4. Nitrogen, N₂
 5. Mixtures of gases, such as air, as long as there is no change of phase.

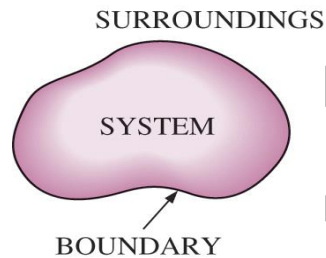
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❖ Phases of A Pure Substance:

- The substances exist in different phases, e.g. at room temperature and pressure, copper is solid and mercury is a liquid.
- It can exist in different phases under variations of condition.
- There are 3 Principal phases,
 1. Solid.
 2. Liquid.
 3. Gas, Each with different molecular structures.

❖ Type of system:

Before discussing types let see the terms:



- A **Thermodynamic system** is defined as a specified region where transfers of mass or energy are to be studied.
- **Boundary** is defined as the real or imaginary surface that separates the system from its surroundings. It may be fixed or moving.
- Everything outside the system is known as **surroundings**.
- When system and surrounding put together is known as **Universe**.
- There are three types of system,

1. **Open system:** In this system, both mass and energy can cross the selected boundary.

Ex. steam turbine, steam nozzle (Fig.4)

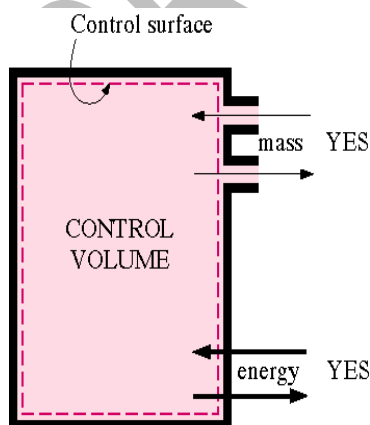


Fig.4

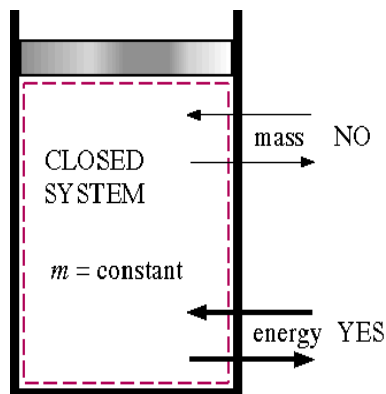


Fig. 5

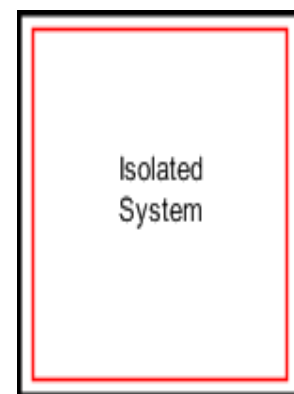


Fig.6

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2. **Closed system:** Here, only transfer of energy but no transfer of mass across the boundary.

Ex. Refrigeration system, closed vessel. (Fig.5)

3. **Isolated system:** An isolated system is that system which does not transfer energy as well as mass with the surrounding. It is purely theoretical system Ex. Thermass. (Fig. 6)

❖ Properties of system:

- Properties of a system defined as any observable or measurable characteristic of a system that is in equilibrium.
- Properties may be intensive or extensive.

1. **Intensive property:** It is defined as the property which does not depend upon the mass of the system. Ex. Pressure, temperature, specific volume, density, specific enthalpy.

2. **Extensive property:** The property which depend on mass of the system, known as Extensive property. Ex. Enthalpy, entropy, total volume.

❖ Thermodynamic state:

- When a system does not undergo any change, all the properties have fixed values. This condition is known as a thermodynamic state.

❖ Process & Cycle:

- When a system is taken from one equilibrium state to another equilibrium state, the change is known as **process**.
- The series of intermediate states through which a system passes during a process is called the **path of the process**.
- If all these intermediate states are equilibrium states, the process is known as quasi equilibrium or **quasi-static process** (Quasi means almost) or reversible process.
- Any process or series of processes whose end states are identical is termed a **cycle**.

❖ Point Function:

- When two properties locate a point on the graph (co-ordinate axes) then those properties are called as **point function**.

Or

- The properties of the system whose change depends on only initial and final state of the system and not on the path followed by the system then such properties are called point function

$$\int_1^2 dv = v_2 - v_1$$

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- It is an exact differential.
- Examples. Internal energy, enthalpy, entropy, Pressure, temperature, volume etc.

❖ Path Function:

- There are certain quantities which cannot be located on a graph by a point but are given by the area on that graph.

Or

- The properties of a system which are depend on path followed by the system and not depend on point or state of the system are called path function.

$$\int_1^2 dQ \neq Q_2 - Q_1$$

- It is an inexact differential. Their change cannot be written as difference between their end states.
- Examples. Heat, work etc.

❖ Thermodynamic work:

- Work is 'something' which appears at the boundary when a system changes its state due to the movement of a part of the boundary under the action of a force. It is denoted by latter 'W'
- Work is said to be done when a force moves through a distance. If a part of the boundary of a system undergoes a displacement under the action of a pressure, the work is done.
- **Sign convention :**
 1. If the work is done by the system on the surroundings, the work is said to be positive.
Ex. An expansion of gas inside the cylinder moves the piston outside.
 2. If the work is done on the system by the surroundings, the work is said to be negative.
Ex. when a force is applied to a piston to compress a fluid.

❖ Heat:

- Heat is 'something' which appears at the boundary when a system changes its state due to a difference in temperature between the system and its surroundings
- It is denoted by the letter Q
- **Sign convention :**
 1. If Heat is received by the system from the surroundings, the quantity is said to be positive.
 2. If Heat is rejected by the system to the surroundings, the quantity is said to be negative.

- ❖ **Potential Energy:** It is the energy possessed by a body at rest condition. Ex. Energy stored in dam water.

$$P.E. = (m * g * h)$$

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- ❖ **Kinetic Energy:** It is the energy possessed by a body when it is in motion. Ex. Energy carried by the ball which is thrown upward.

$$K.E. = \frac{1}{2}mv^2$$

- ❖ **Internal Energy:** It is the algebraic sum of internal kinetic energy and the internal potential energy of its molecules. As it depends on mass of the system, hence it is extensive property. And denoted by letter u .

- ❖ **Flow Work:** The flow energy caused the flow of $V_s \text{ m}^3$ of gas against a pressure P of N/m^2 is known as flow work. And given by,

$$F.W. = (P * V_s) \text{ N-m}$$

- ❖ **Enthalpy:** It is the algebraic sum of the internal energy and flow work. It is the total heat content of the system. Denoted by letter H or h .

$$H = (u + PV_s) \text{ KJ/Kg}$$

Change in enthalpy is written as, $\Delta H = H_2 - H_1 = mc_p(t_2 - t_1)$

- ❖ **Entropy:**

- The quantity $\frac{\delta Q}{T}$ is a function of the initial and final states of the system and is independent of the path of the process. Hence it represents a property of the system. This property is known as the entropy.
- Entropy is a function of a quantity of heat which shows the possibility of conversion of that heat into work. The increase in entropy is small when heat is added at a high temperature and is greater when heat addition is made at a lower temperature.
- It denoted by letter S . and change in entropy is written as,

$$\Delta S = \frac{\delta Q}{T}$$

- Unit of entropy – KJ/Kg°k

- ❖ **Zeroth Law of Thermodynamics:**

It states that “If two bodies are separately in thermal equilibrium with a third body, then these are also in thermal equilibrium with each other.”

From fig. if body A and B are in thermal equilibrium with a body C, then bodies A and B must be thermal equilibrium with each other.



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Ex. The measuring of temperature of any liquid by a thermometer.

❖ Statements of First law of thermodynamics:

1. **First law for cyclic process:** It states that whenever a system under goes a cyclic change, the algebraic sum of work transfer is proportional to algebraic sum of heat transfer.

$$\oint dQ = \oint dW$$

2. **First law for closed cycle for non cyclic process:** it states that if system under goes a process during with both heat transfer and work transfer involved, the net energy transferred will be stored within the system.

$$\Delta U = dQ - dW$$

3. **Law of conservation of energy:** It states that “The energy can neither be created nor destroyed though it can be transferred from one form to another.”
4. No machine can produce energy without corresponding expenditure of energy, i.e., it is impossible to construct a perpetual motion machine of first kind.

❖ Irreversibility:

- Is the process
- It is defined as the difference between reversible work and useful work. It is denoted by

$$I = (W_{rev} - W_{useful}) \text{ or } I = (W_{useful} - W_{rev})$$

- The irreversibility is equivalent to energy destroyed. If reversible work and useful work are identical then irreversibility to zero.

$$W_{rev} \geq W_{useful} \rightarrow \text{work producing device}$$

$$W_{rev} \leq W_{useful} \rightarrow \text{work consuming device}$$

- The smaller the irreversibility greater the work will be produced. Hence the performance can be improved by minimizing the irreversibility.
- **The factors that cause a process to be irreversible are Examples:**
 1. Friction.
 2. Unrestrained expansion.
 3. Mixing of two gases i.e. combustion.
 4. Heat transfer across a finite temperature difference.
 5. Spontaneous chemical reactions.
 6. Expansion or Compression with finite pressure difference.
 7. Mixing of matter at different states.

❖ Steady Flow Energy Equation:

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The Steady Flow Energy Equation (SFEE) is used for open systems to determine the total energy flows.

The following assumptions are made in the system

1. The mass flow rate through the system remains constant.
2. In the analysis only potential, kinetic and flow energies are considered.
3. The state of fluid at any point remains constant with time.
4. The rate of heat transfer & work transfer is constant.
5. There is no change in chemical composition of the system.

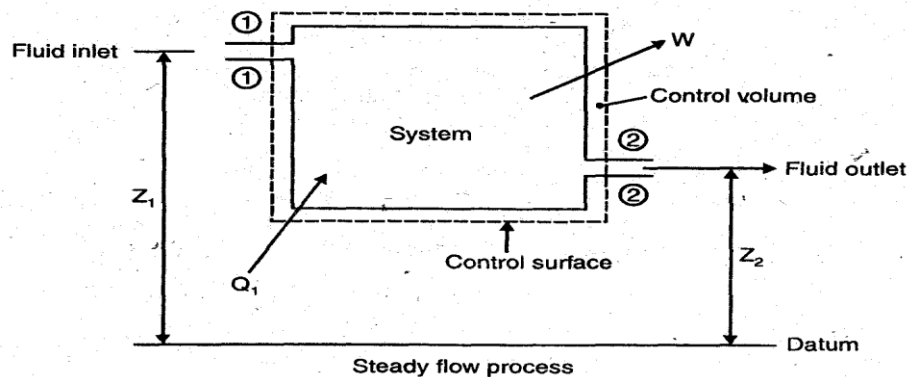


Fig. Shows a flow process for an open system with control volume.

Let $P_1, P_2, V_1, V_2, V_{s1}, V_{s2}, h_1, h_2, u_1$ & u_2 are pressure, velocity, specific volume, enthalpy, internal energy at inlet (1) and outlet (2) condition.

Z_1, Z_2 are elevation at inlet and outlet about datum line. W is work done and Q is heat transfer.

$$\frac{dQ}{dt} \rightarrow \text{heat transfer rate} \quad \frac{dW}{dt} \rightarrow \text{work transfer rate}$$

$$m(\text{energy at inlet}) + m(\text{flow energy at inlet}) + \frac{dQ}{dt} = m(\text{energy at exit}) + m(\text{flow energy at exit}) + \frac{dW}{dt}$$

By considering unit mass $m = 1$ & energy = (K.E. + P.E. + I.E.)

$$(K.E. + P.E. + I.E.)_1 + (P_1 V_{s1}) + \frac{dQ}{dt} = (K.E. + P.E. + I.E.)_2 + (P_2 V_{s2}) + \frac{dW}{dt}$$

$$\frac{1}{2} V_1^2 + Z_1 g + u_1 + (P_1 V_{s1}) + \frac{dQ}{dt} = \frac{1}{2} V_2^2 + Z_2 g + u_2 + (P_2 V_{s2}) + \frac{dW}{dt}$$

By keeping $\frac{dQ}{dt} \rightarrow q$ $\frac{dW}{dt} \rightarrow w$ & $h_1 = u_1 + P_1 V_{s1}$ $h_2 = u_2 + P_2 V_{s2}$

$$\frac{1}{2} V_1^2 + Z_1 g + h_1 + q = \frac{1}{2} V_2^2 + Z_2 g + h_2 + w$$

$$\frac{(V_2^2 - V_1^2)}{2} + (Z_2 - Z_1)g + (h_2 - h_1) + w = q$$

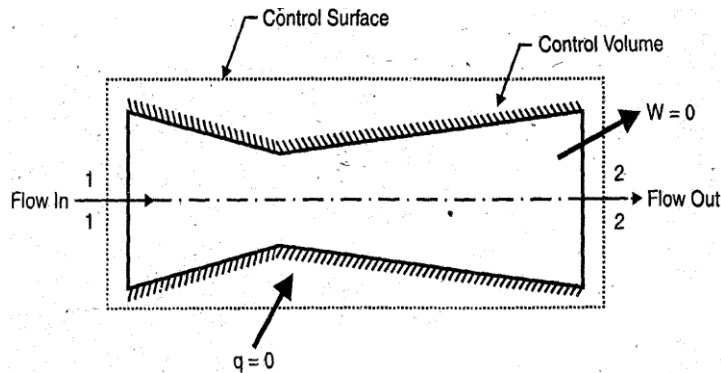
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$$q - w = \frac{(v_2^2 - v_1^2)}{2} + (z_2 - z_1)g + (h_2 - h_1)$$

❖ Application of SFEE:

1. Nozzle:

It is a passage of varying cross-sectional area by means of which pressure energy of flowing fluid is converted into kinetic energy. And it is used for producing high velocity jet.



There is no work as well as heat transfer and nozzle is kept horizontal

$$\therefore z_1 = z_2 = 0, \quad q = 0 \text{ \& } w = 0$$

$\therefore$ SFEE reduces to,

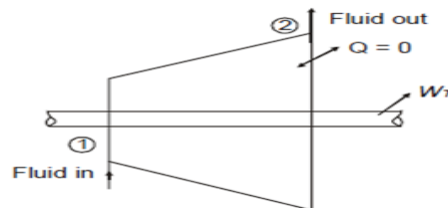
$$0 = \frac{(v_2^2 - v_1^2)}{2} + (h_2 - h_1)$$

$$v_2 = \sqrt{2(h_1 - h_2) + v_1^2}$$

$$v_2 = \sqrt{2 C_p(t_1 - t_2) + v_1^2}$$

2. Steam Turbine:

Steam turbine converts the heat energy of steam into useful work.



By applying SFEE,

$$q - w = \frac{(v_2^2 - v_1^2)}{2} + (z_2 - z_1)g + (h_2 - h_1)$$

In case of turbine, change in K.E. & P.E. are negligible and heat transfer $Q=0$

$$\therefore -w = (h_2 - h_1)$$

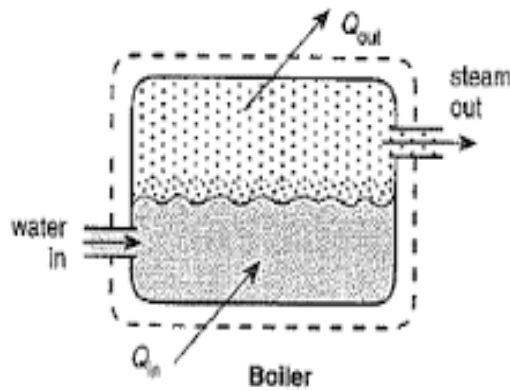
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$$\therefore w = (h_1 - h_2)$$

Thus enthalpy drop is converted into useful work.

3. Boiler:

Here, during combustion process, chemical energy of the fuel is converted into heat energy. This heat energy is utilized to convert water into steam.



Applying steady flow energy equation,

$$q - w = \frac{(V_2^2 - V_1^2)}{2} + (z_2 - z_1)g + (h_2 - h_1)$$

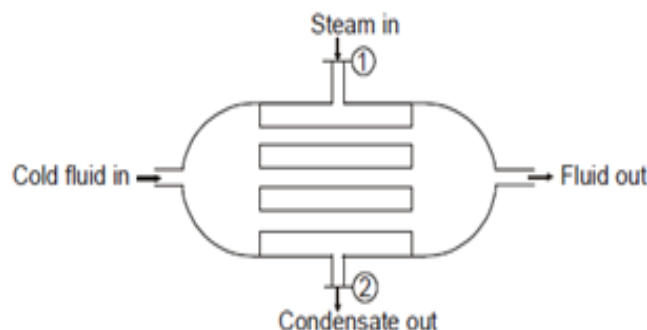
In case of boiler, change in K.E. & P.E. are negligible and no work is done $w = 0$

$$\therefore q = (h_2 - h_1)$$

Thus heat supplied is utilized to increase enthalpy.

4. Condenser:

The condenser is used to condense the steam by absorbing heat from steam with the help of cooling water.



Applying steady flow energy equation,

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$$q - w = \frac{(V_2^2 - V_1^2)}{2} + (z_2 - z_1)g + (h_2 - h_1)$$

In case of condenser, no change in K.E., P.E. is negligible and no work is done $w = 0$
As q is taken as **negative** because heat is rejected by the system.

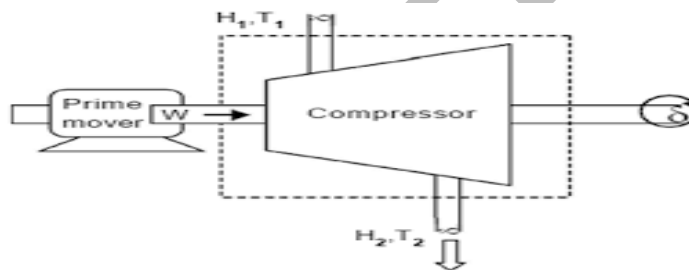
$$\therefore -q = (h_2 - h_1)$$

$$\therefore q = (h_1 - h_2)$$

Thus heat supplied is utilized to increase enthalpy.

5. Reciprocating Compressor:

The reciprocating compressor sucks in air from atmosphere and supplies at a considerable higher pressure in small quantities. The reciprocating compressor can be considered as steady flow system provided the control volume includes the receiver which reduces the fluctuations of flow considerably.



Applying steady flow energy equation,

$$q - w = \frac{(V_2^2 - V_1^2)}{2} + (z_2 - z_1)g + (h_2 - h_1)$$

In case of compressor, change in K.E. & P.E. is negligible

Here, work is done on the system, hence it is negative and heat is transfer from the system hence negative.

$$\therefore -q - (-w) = (h_2 - h_1)$$

$$-q + w = (h_2 - h_1)$$

$$\therefore w = (h_2 - h_1) + q$$

❖ Limitations of First Law of Thermodynamics:

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1. First law states that work transfer is equal to heat transfer and it does not provide any restriction on the direction of the process.
2. First law of thermodynamics has not provided any information about the conditions in which conversion of heat energy in work energy is possible.
3. According to this statement the potential energy can be converted into kinetic energy and kinetic energy can be converted into potential energy, but in natural practice this do not happen.
4. Ex. A car driver up the hill, the level of fuel in the tank drop. If this process reverse i.e. car down the hill, the fuel consume cannot be refill in tank.
5. Hence from above example the reversal of process is not true without the addition of external work.

❖ **Second Law of Thermodynamics:**

1. Kelvin-Planck Statement:

It is impossible to construct a heat engine, which works in a cyclic process whose sole effect is to convert all the heat supplied into an equivalent amount of work. (Which gives 100% efficiency)

2. Clausius Statement:

It is impossible to construct a heat pump to work in cyclic process whose sole effect is the transfer of heat from a body at lower temperature to higher temperature.

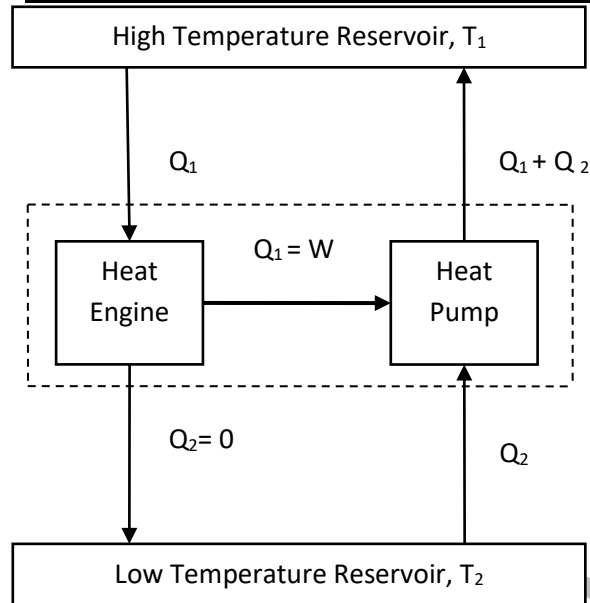
❖ **Equivalence of Kelvin- plank And Clausius Statement:**

Although the Kelvin Plank and Clausius statements appears to be different, but they are equivalent (same). The violation of one means the violation of other. This can be proved very easily as follow.

• **Violation on Kelvin Plank Statement:**

Consider the heat pump which violates the Kelvin Plank statement by absorbing heat Q_1 from single reservoir and producing equal amount of work $W = Q_1$. The work delivered by this heat engine is used to drive the heat pump which takes the heat Q_2 from low temperature reservoir and delivers $Q_1 + Q_2$ amount of heat to high temperature reservoir by consuming the work $W = Q_1$ as shown.

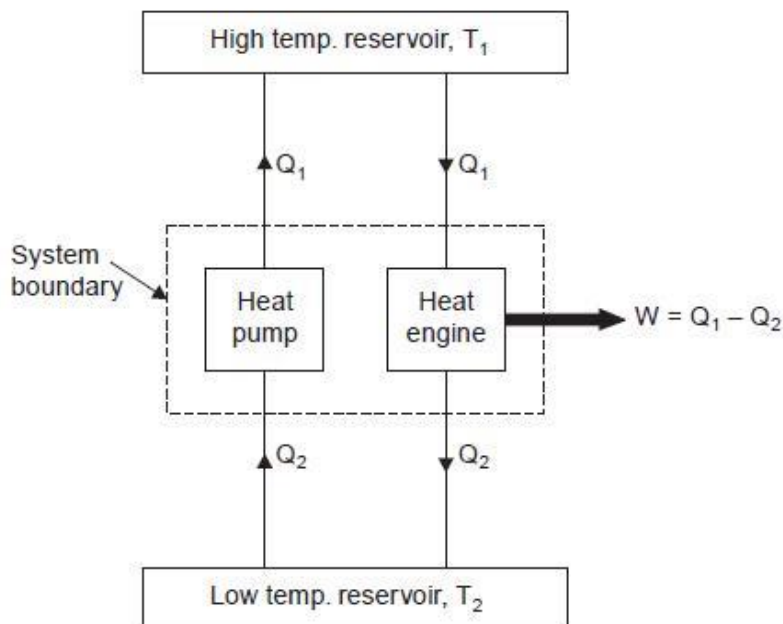
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Thus if we consider the combined system of heat engine and heat pump, then it acts like a heat pump transferring the Q_2 amount of heat from low temperature to high temperature reservoir without consuming any external work. Thus it violates the Clausius statement.

- Violation on Clausius Statement:**

Consider Heat pump, which violates the clausius statement by transferring Q_2 amount of heat from low temperature to high temperature body without consumption of work ($w = 0$). Let assume the heat engine working between the same thermal reservoir which absorb Q_1 amount heat (which is greater than Q_2) and delivers work $W = Q_1 - Q_2$ and reject the Q_2 amount of heat to low temperature reservoir as shown in fig.



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Thus if we consider the combined system of heat engine and heat pump, then it acts like heat engine which absorbs the $Q_1 - Q_2$ amount of heat and develops the equivalent amount of work. Thus it acts like heat engine exchanging heat single reservoir and producing an equal amount of work. This violates the Kelvin statement

❖ PPM-1 and PPM-2 Kind:

1. PPM-1:

- a machine that does not follow the first law of thermodynamics could be termed as perpetual motion machine of first kind i.e. PMM1
- Such machines will produce the energy by itself and as we know that according to the law of energy conservation, energy could not be created or destroyed but could be converted from one form of energy to other form of energy.(Fig. a)

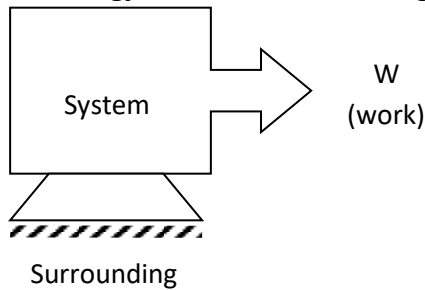


Fig. a

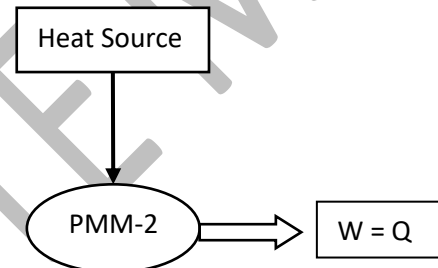


Fig. b

2. PPM-2:

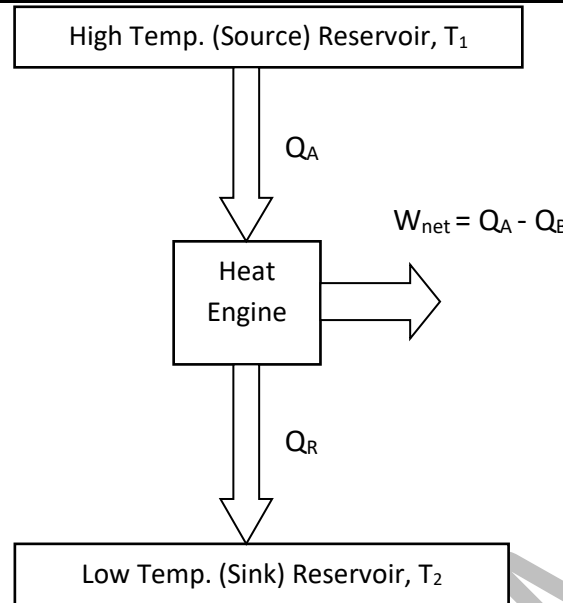
- Perpetual motion machines of second kind are those machines that violate the second law of thermodynamics because such machines will absorb continuously heat energy from a single thermal reservoir and will convert the absorbed heat energy completely in to work energy.
- Such machines will have efficiency 100%, it is not possible. (Fig. b)

❖ Applications of Second Law Of Thermodynamics:

1. Heat Engine:

- It is a device which converts chemical energy of fuel into thermal energy to perform some useful work, when operating in a cyclic process.

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Where,

Q_A = heat added to system.

Q_R = Heat Rejected to system.

W_{net} = Net work done by system.

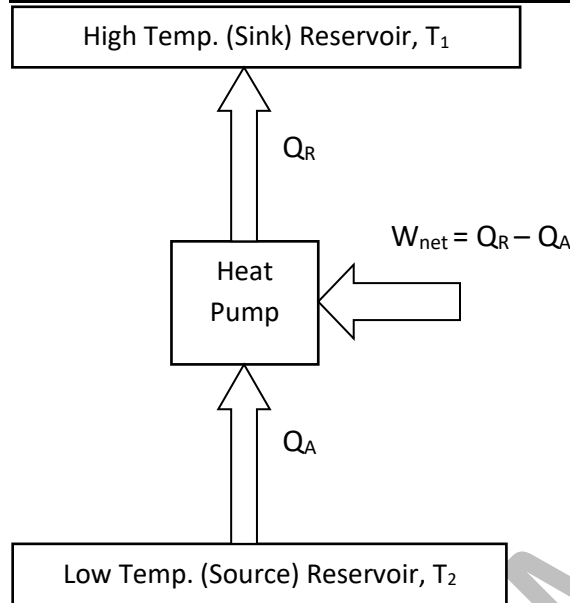
$$\begin{aligned} \text{Thermal Efficiency} &= \frac{\text{Net work output}}{\text{Heat Input}} = \frac{W_{net}}{Q_A} \\ &= \frac{Q_A - Q_B}{Q_A} = \frac{T_1 - T_2}{T_1} \end{aligned}$$

∴ Thermal efficiency of heat engine operating between two temperatures is always less than unity.

2. Heat Pump:

- It is a Device which works in a cycle delivers heat from low temperature to high temperature, for the heating purpose.
- The efficiency of heat pump is replaced by the term Coefficient of Performance (COP), just to avoid confusion.
- Here, the purpose of heat pump is to take the heat from low temperature (T_2) and reject it to a room at temperature of (T_1) for heating purpose.

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$$\therefore \text{Thermal Efficiency} = \frac{\text{heat Rejected}}{\text{Net work output}} = \frac{Q_R}{W_{net}}$$

$$= \frac{Q_R}{Q_R - Q_A} = \frac{T_1}{T_1 - T_2}$$

3. Refrigerator:

- If The purpose of Heat Pump is to extract the heat from the source which is at low temperature (T_2) and rejected to the sink at high temperature (T_1)
- Here, the function of Heat Pump is to cool the source i.e. the cooling of room hence that heat pump is known as Refrigerator. (Ref. same fig. for heat pump)

$$\therefore \text{Thermal Efficiency} = \frac{\text{heat Absorbed}}{\text{Net work output}} = \frac{Q_A}{W_{net}}$$

$$= \frac{Q_A}{Q_R - Q_A} = \frac{T_2}{T_1 - T_2}$$

❖ Prove that $(C.O.P.)_{\text{heat pump}} = (C.O.P.)_{\text{refri}} + 1$

Solution: T_1 = High Temp. T_2 = Low Temp.

$$(C.O.P.)_{\text{ref}} + 1 = \frac{T_2}{T_1 - T_2} + 1$$

$$= \frac{T_2 + T_1 - T_2}{T_1 - T_2} = \frac{T_1}{T_1 - T_2}$$

$$(C.O.P.)_{\text{ref}} + 1 = (C.O.P.)_{\text{heat pump}}$$

❖ Difference between:

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Sr. no.	work	Heat
1	It is energy in transition across boundary of system due to travel of force	It is energy in transition across boundary of system due to temperature difference
2	Thermodynamic work, mechanical work, shaft work etc are form of work	Enthalpy, entropy are forms of it
3	Work supplied to system is negative. Work done by the system is positive by convention.	Heat rejected by system is negative. Heat supplied to system is positive by convention.
4	Unit is N-m	Unit is Joule

Sr. no	Heat pump	Refrigerator
1	it works between hot body temp and atmospheric temperature.	It works between cold body temp and atmospheric temperature.
2	Heat pump used for heating purpose.	Refrigerator used for cooling purpose.
3	$\eta = \frac{Q_R}{Q_R - Q_A}$	$\eta = \frac{Q_A}{Q_R - Q_A}$
4	COP of heat pump is greater than COP of refrigerator by unity.	COP is less
5	In case of HP atmosphere acts as a cold body	In case of refrigerator atmosphere acts as a hot body.

Sr. no	Open System	Closed System
1	In this system, Both mass & energy may cross the boundaries.	Only energy transfer across the boundaries. not the mass
2	Mass of the system do not remains constant.	Mass of the system remains constant.
3	It can be explain with concept of control volume and control surface.	It can be explain with the concept of boundaries.

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4	Ex. Steam turbine, compressor, condenser pump etc.	Ex. Piston & cylinder without valve.
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Sr. no	Heat Engine	Heat Pump/ Refrigerator
1	It works on Kelvin plank's statement of second law of thermodynamics.	It works on Clausius statement of second law of thermodynamics.
2	It is a work developing device.	It is a work absorbing device.
3	Its performance measured by term known as Efficiency.	Its performance measured by term Coefficient of Performance.
4	It converts heat energy into useful work.	It transfers heat from low temperature body to high temp. body
5	$\eta = \frac{Q_A - Q_R}{Q_A}$	$\eta = \frac{Q_A}{Q_R - Q_A}$